# Chapter 4: Probability Theory and Distributions

Master in Computer Science (M1)

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# 1 Introduction

Probability theory provides the mathematical language used to describe uncertainty. In many real situations, the result of an experiment cannot be predicted with certainty before it is observed. For example, we may not know whether an email is spam, whether a customer will click on an advertisement, how many users will visit a website, or what value a sensor will record at a given time. Probability allows us to model these situations in a rigorous way.

This chapter introduces the fundamental notions needed to understand probabilistic reasoning and probability distributions. We begin with the concepts of sample space and events, which allow us to describe all possible outcomes of a random experiment and the specific situations of interest. We then study the main operations on events, such as union, intersection, complement, and difference. These operations are important because many probability problems are naturally expressed using set relationships.

After that, we introduce conditional probability, which describes how the probability of an event changes when new information is available. This leads naturally to the notion of independence, where the occurrence of one event does not influence the probability of another. These ideas are essential in statistics and machine learning, especially when studying relationships between variables, features, and classes.

A central result of this chapter is Bayes' theorem. Bayes' theorem gives a mathematical rule for updating probabilities when new evidence is observed. It is widely used in machine learning, for example in spam filtering, medical diagnosis, Bayesian inference, and classification methods such as the Naive Bayes classifier.

The second part of the chapter is devoted to random variables and probability distributions. A random variable transforms the outcomes of a random experiment into numerical values. We distinguish between discrete random variables, which take countable values, and continuous random variables, which take values in intervals of the real line.

For discrete random variables, we study important probability laws such as the discrete uniform distribution, Bernoulli distribution, binomial distribution, and Poisson distribution. These distributions are useful for modeling counts, binary outcomes, rare events, and repeated trials.

For continuous random variables, we introduce the probability density function and the cumulative distribution function. We then study several continuous distributions, with particular emphasis on the normal distribution, which plays a fundamental role in statistics, data science, and machine learning.

Finally, we study the Central Limit Theorem. This theorem explains why the normal distribution appears so frequently in practice. It shows that, under suitable conditions, the sample mean of a large number of independent observations is approximately normally distributed. This result is one of the foundations of statistical inference, confidence intervals, hypothesis testing, and many data analysis methods.

The objective of this chapter is therefore not only to present formulas, but also to build intuition about randomness, uncertainty, and probabilistic modeling. These tools will be used later to understand statistical learning methods and to interpret results obtained from real data.

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# 2 Probability Spaces and Events

Probability theory starts with the description of all possible outcomes of a random experiment. Before assigning probabilities, we must clearly define what can happen and which events are of interest.

## 2.1 Sample Space

A **random experiment** is an experiment whose result cannot be predicted with certainty before it is performed.
The set of all possible outcomes is called the **sample space** and is denoted by

$$\Omega.$$

Examples of sample spaces are:

$$\Omega = \{H, T\}$$

for tossing a coin once,

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

for rolling a die, and

$$\Omega = [-50, 60]$$

for measuring a temperature between $-50°C$ and $60°C$.

## 2.2 Events

An **event** is a subset of the sample space $\Omega$.
If the outcome of the experiment belongs to this subset, we say that the event has occurred.

For example, when rolling a die, the sample space is

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

The event of obtaining an even number is

$$A = \{2, 4, 6\}.$$

The event of obtaining a number greater than 3 is

$$B = \{4, 5, 6\}.$$

Two special events are:

$$\Omega$$

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which is called the certain event, and

$$\emptyset$$

which is called the impossible event.

## 2.3 Operations on Events

Since events are sets, we can combine them using classical set operations.

- The union $A \cup B$ is the event that $A$ occurs or $B$ occurs.
- The intersection $A \cap B$ is the event that both $A$ and $B$ occur.
- The complement $A^c$ is the event that $A$ does not occur.
- The difference $A - B$ is the event that $A$ occurs but $B$ does not occur.

For the previous example,

$$A = \{2, 4, 6\}, \qquad B = \{4, 5, 6\}.$$

Then

$$A \cup B = \{2, 4, 5, 6\},$$

$$A \cap B = \{4, 6\},$$

and

$$A^c = \{1, 3, 5\}.$$

**Important Formulas**

For two events $A$ and $B$,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If $A$ and $B$ are disjoint, then

$$A \cap B = \emptyset$$

and therefore

$$P(A \cup B) = P(A) + P(B).$$

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## 2.4 De Morgan's Laws

The complement of a union and the complement of an intersection are given by De Morgan's laws:

$$(A \cup B)^c = A^c \cap B^c$$

and

$$(A \cap B)^c = A^c \cup B^c.$$

These identities are useful in probability, logic, and computer science.

# 3 Conditional Probability and Independence

In many real situations, the probability of an event changes when new information becomes available. Conditional probability allows us to update a probability when we know that another event has occurred.

## 3.1 Conditional Probability

Let $A$ and $B$ be two events such that

$$P(B) > 0.$$

The conditional probability of $A$ given $B$ is defined by

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

This means that once we know that $B$ has occurred, the sample space is reduced to $B$, and we measure how often $A$ also occurs inside $B$.

**Example 3.1.** *Suppose we study emails and define the following events:*

$$S = \{\textit{the email is spam}\}$$

*and*

$$O = \{\textit{the email contains the word "offer"}\}.$$

*Assume that*

$$P(S) = 0.3, \qquad P(O) = 0.2, \qquad P(S \cap O) = 0.15.$$

*The probability that an email is spam given that it contains the word "offer" is*

$$P(S \mid O) = \frac{P(S \cap O)}{P(O)} = \frac{0.15}{0.20} = 0.75.$$

*Thus, knowing that the email contains the word "offer" increases the probability of spam from* 0.3 *to* 0.75.

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## 3.2 Independence

Two events $A$ and $B$ are called **independent** if the occurrence of one event does not change the probability of the other.
Mathematically,

$$P(A \cap B) = P(A)P(B).$$

Equivalently, if $P(B) > 0$, then

$$P(A \mid B) = P(A).$$

**Example 3.2.** *Consider two tosses of a fair coin.*
*Let*

$$A = \{\textit{the first toss is heads}\}$$

*and*

$$B = \{\textit{the second toss is heads}\}.$$

*Then*

$$P(A) = \frac{1}{2}, \qquad P(B) = \frac{1}{2},$$

*and*

$$P(A \cap B) = \frac{1}{4}.$$

*Since*

$$P(A \cap B) = \frac{1}{4} = \frac{1}{2} \times \frac{1}{2} = P(A)P(B),$$

*the events $A$ and $B$ are independent.*

**Example 3.3.** *Consider drawing two cards from a deck without replacement.*
*Let*

$$A = \{\textit{the first card is an ace}\}$$

*and*

$$B = \{\textit{the second card is an ace}\}.$$

*If the first card is an ace, then only* 3 *aces remain among* 51 *cards. Hence*

$$P(B \mid A) = \frac{3}{51}.$$

*However,*

$$P(B) = \frac{4}{52}.$$

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*Since*

$$P(B \mid A) \neq P(B),$$

*the events are not independent.*

# 4 Bayes' Theorem

Bayes' theorem is one of the most important formulas in probability theory. It allows us to reverse conditional probabilities and update our belief about an event after observing new evidence.

This idea is fundamental in machine learning, especially in Bayesian models, spam filtering, medical diagnosis, and classification problems.

**Bayes' Theorem**

Let $A$ and $B$ be two events such that

$$P(A) > 0.$$

Then

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}.$$

**Important Formulas**

Bayes' theorem can be interpreted as

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}.$$

Here,

$$P(B)$$

is the prior probability,

$$P(A \mid B)$$

is the likelihood,

$$P(A)$$

is the evidence, and

$$P(B \mid A)$$

is the posterior probability.

**Example 4.1.** *Suppose we want to predict whether it will rain.*

*Let*

$$R = \{\textit{it rains today}\}$$

*and*

$$C = \{\textit{the sky is cloudy in the morning}\}.$$

*Assume that*

$$P(R) = 0.3, \qquad P(C) = 0.4, \qquad P(C \mid R) = 0.8.$$

*We want to compute*

$$P(R \mid C).$$

*Using Bayes' theorem,*

$$P(R \mid C) = \frac{P(C \mid R)P(R)}{P(C)}.$$

*Therefore,*

$$P(R \mid C) = \frac{0.8 \times 0.3}{0.4} = 0.6.$$

*Thus, after observing that the sky is cloudy, the probability of rain increases from* 0.3 *to* 0.6*.*

**Example 4.2.** *In a medical diagnosis problem, let*

$$D = \{\textit{the patient has a disease}\}$$

*and*

$$T = \{\textit{the test is positive}\}.$$

*Bayes' theorem gives*

$$P(D \mid T) = \frac{P(T \mid D)P(D)}{P(T)}.$$

*This formula helps us compute the probability that a patient really has the disease after receiving a positive test result. It is important because a positive test may occur both for sick patients and for healthy patients due to false positives.*

# 5 Discrete Random Variables

In many experiments, the outcome can be represented by a numerical value. For example, the number of defective products in a batch, the number obtained when rolling a die, or the number of infected individuals in a population.

To model such quantities, we introduce the notion of a discrete random variable, which associates a real number with each outcome of a random experiment.

Consider a random experiment with sample space

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$$\Omega = \{\omega_1, \omega_2, \ldots, \omega_n\}.$$

Let $P$ be a probability measure defined on $\Omega$.

A **discrete random variable** is a function

$$X : \Omega \longrightarrow \mathbb{R}$$

such that

$$\omega_i \longmapsto X(\omega_i) = x_i.$$

The random variable $X$ takes values in the set

$$\{x_1, x_2, \ldots, x_p\}.$$

The probability distribution of $X$ is given by

$$p_i = P(X = x_i),$$

satisfying

$$p_i \geq 0, \qquad \forall i,$$

and

$$\sum_i p_i = 1.$$

## 5.1 Expectation

The expectation, also called the mean or expected value, represents the average value that would be observed if the experiment were repeated a large number of times. It can be viewed as the center of gravity of the probability distribution.

The expectation of $X$ is defined by

$$E(X) = \sum_i x_i \, P(X = x_i).$$

## 5.2 Variance

While the expectation measures the central tendency of a random variable, the variance measures its dispersion around the mean. A small variance indicates that the values are concentrated around the expectation, whereas a large variance indicates greater variability.

The variance of $X$ is

$$V(X) = E(X^2) - [E(X)]^2,$$

where

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$$E(X^2) = \sum_i x_i^2 \, P(X = x_i).$$

**Important Formulas**

For a discrete random variable $X$,

$$E(X) = \sum_x xP(X = x)$$

and

$$V(X) = E(X^2) - [E(X)]^2.$$

These formulas will be used throughout the chapter to compute the mean and variability of discrete probability distributions.

**Example 5.1.** *An urn contains two red balls $R_1, R_2$ and three black balls $B_1, B_2, B_3$. Two balls are selected simultaneously without replacement. Let $X$ denote the number of red balls obtained.*

$$X \in \{0, 1, 2\}.$$

*The probabilities are*

$$P(X = 0) = P(\textit{two black balls}) = \frac{\binom{3}{2}}{\binom{5}{2}} = \frac{3}{10},$$

$$P(X = 1) = P(\textit{one red and one black}) = \frac{\binom{2}{1}\binom{3}{1}}{\binom{5}{2}} = \frac{6}{10},$$

$$P(X = 2) = P(\textit{two red balls}) = \frac{\binom{2}{2}}{\binom{5}{2}} = \frac{1}{10}.$$

*Hence,*

| $x$ | 0 | 1 | 2 |
|---|---|---|---|
| $P(X = x)$ | $\frac{3}{10}$ | $\frac{6}{10}$ | $\frac{1}{10}$ |

*The expectation is*

$$E(X) = 0 \cdot \frac{3}{10} + 1 \cdot \frac{6}{10} + 2 \cdot \frac{1}{10} = \frac{8}{10} = 0.8.$$

*Furthermore,*

$$E(X^2) = 0^2 \cdot \frac{3}{10} + 1^2 \cdot \frac{6}{10} + 2^2 \cdot \frac{1}{10} = 1.$$

*Therefore,*

$$V(X) = E(X^2) - [E(X)]^2 = 1 - (0.8)^2 = 0.36.$$

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## 5.3 Discrete Uniform Distribution

In this section, we study one of the simplest discrete probability distributions: the discrete uniform distribution. It is used when all possible values have the same probability of occurring.

A random variable $X$ has a **discrete uniform distribution** on the set $\{1, 2, \ldots, n\}$, where $n \in \mathbb{N}^*$, if

$$P(X = 1) = P(X = 2) = \cdots = P(X = n) = \frac{1}{n}.$$

We write

$$X \sim \mathcal{U}(\{1, 2, \ldots, n\}).$$

**Important Formulas**

If $X \sim \mathcal{U}(\{1, 2, \ldots, n\})$, then

$$E(X) = \frac{n+1}{2}$$

and

$$V(X) = \frac{n^2 - 1}{12}.$$

**Example 5.2.** *A die is rolled once. Let $X$ be the number obtained.*
*Then*

$$X \in \{1, 2, 3, 4, 5, 6\}$$

*and*

$$P(X = 1) = P(X = 2) = \cdots = P(X = 6) = \frac{1}{6}.$$

*Therefore,*

$$X \sim \mathcal{U}(\{1, 2, 3, 4, 5, 6\}).$$

*The expectation is*

$$E(X) = \frac{6+1}{2} = 3.5.$$

*The variance is*

$$V(X) = \frac{6^2 - 1}{12} = \frac{35}{12}.$$

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## 5.4 Bernoulli Distribution

The Bernoulli distribution is used to model an experiment with only two possible outcomes: success or failure. It is one of the most important distributions because it is the building block of many other probability laws, such as the binomial distribution.

Consider a random experiment with sample space

$$\Omega = \{S, \overline{S}\},$$

where $S$ denotes success and $\overline{S}$ denotes failure.

Let

$$P(S) = p, \qquad P(\overline{S}) = q = 1 - p.$$

A random variable $X$ has a **Bernoulli distribution** with parameter $p$ if

$$X = \begin{cases} 1, & \text{if success occurs,} \\ 0, & \text{if failure occurs.} \end{cases}$$

Thus,

$$P(X = 1) = p, \qquad P(X = 0) = 1 - p.$$

We write

$$X \sim \mathcal{B}(p).$$

**Important Formulas**

If $X \sim \mathcal{B}(p)$, then

$$E(X) = p$$

and

$$V(X) = p(1 - p).$$

**Example 5.3.** *Suppose that the percentage of people affected by a disease is* 0.008.

*Let $X$ describe the state of a randomly selected person:*

$$X = \begin{cases} 1, & \textit{if the person is affected,} \\ 0, & \textit{if the person is not affected.} \end{cases}$$

*Then*

$$X \sim \mathcal{B}(p), \qquad p = P(X = 1) = 0.008.$$

*The expectation is*

$$E(X) = p = 0.008.$$

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*The variance is*

$$V(X) = p(1-p) = 0.008(1-0.008) = 0.007936.$$

## 5.5 Binomial Distribution

The Bernoulli distribution models one experiment with two possible outcomes: success or failure. When this experiment is repeated several times independently, we are often interested in the number of successes obtained. This situation is modeled by the binomial distribution.

Let $X$ be the number of successes in $n$ independent Bernoulli trials, where the probability of success is $p$ and the probability of failure is $q = 1 - p$.
Then $X$ follows a **binomial distribution** with parameters $n$ and $p$. We write

$$X \sim \mathcal{B}(n, p).$$

The possible values of $X$ are

$$X \in \{0, 1, \ldots, n\}.$$

For $k = 0, 1, \ldots, n$, the probability distribution is

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

**Important Formulas**

If $X \sim \mathcal{B}(n, p)$, then

$$E(X) = np$$

and

$$V(X) = np(1-p).$$

**Example 5.4.** *Suppose that the probability that a person is affected by a disease is*

$$p = 0.001.$$

*We select* 20 *people independently.*
*Let* $X$ *be the number of affected people among the* 20 *selected people. Then*

$$X \sim \mathcal{B}(20, 0.001).$$

*Thus, for* $k = 0, 1, \ldots, 20$,

$$P(X = k) = \binom{20}{k} (0.001)^k (0.999)^{20-k}.$$

*The expectation is*

$$E(X) = 20 \times 0.001 = 0.02.$$

*The variance is*

$$V(X) = 20 \times 0.001 \times 0.999 = 0.01998.$$

**Example 5.5.** *Using the previous example, we calculate the probability that fewer than two people are affected.*

$$P(X < 2) = P(X = 0) + P(X = 1).$$

*Therefore,*

$$P(X < 2) = \binom{20}{0}(0.001)^0(0.999)^{20} + \binom{20}{1}(0.001)^1(0.999)^{19}.$$

*Hence,*

$$P(X < 2) \approx 0.9998.$$

## 5.6 Poisson Distribution

The Poisson distribution is used to model the number of occurrences of an event during a fixed period of time or in a fixed region. It is especially useful for rare events.

Typical examples include the number of failures per year, the number of phone calls per hour, or the number of accidents per day.

A random variable $X$ follows a **Poisson distribution** with parameter $\lambda > 0$ if

$$P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k \in \mathbb{N}.$$

We write

$$X \sim \mathcal{P}(\lambda).$$

Important Formulas

If $X \sim \mathcal{P}(\lambda)$, then

$$E(X) = \lambda$$

and

$$V(X) = \lambda.$$

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## How to Identify a Poisson Distribution

A random variable can be modeled by a Poisson distribution in the following cases:

1. The exercise explicitly states that the variable follows a Poisson distribution.
2. The variable counts the number of occurrences of an event during a fixed unit of time.
3. The variable follows a binomial distribution with $n$ large and $p$ small. In practice, if

$$n > 30, \qquad p < 0.1, \qquad np < 5,$$

then the binomial distribution can be approximated by a Poisson distribution:

$$\mathcal{B}(n, p) \approx \mathcal{P}(np).$$

Thus,

$$P(X = k) \approx e^{-np}\frac{(np)^k}{k!}.$$

**Example 5.6.** *The average number of failures of a component during one year is* 4.

*Let $X$ be the number of failures during one year. Since $X$ counts the number of occurrences during a fixed unit of time, we model it by*

$$X \sim \mathcal{P}(4).$$

*Therefore,*

$$P(X = k) = e^{-4}\frac{4^k}{k!}, \qquad k \in \mathbb{N}.$$

*Moreover,*

$$E(X) = 4$$

*and*

$$V(X) = 4.$$

**Example 5.7.** *Using the previous example, calculate the probability that at least two failures occur during one year.*

$$P(X \geq 2) = 1 - P(X < 2).$$

*Since*

$$P(X < 2) = P(X = 0) + P(X = 1),$$

*we get*

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$$P(X \geq 2) = 1 - (P(X = 0) + P(X = 1)).$$

*Thus,*

$$P(X \geq 2) = 1 - e^{-4}\left(\frac{4^0}{0!} + \frac{4^1}{1!}\right).$$

*Therefore,*

$$P(X \geq 2) = 1 - 5e^{-4}.$$

**Example 5.8.** *Suppose that* $4\%$ *of a population carries a genetic trait. A sample of* $100$ *individuals is selected.*

*Let* $X$ *be the number of individuals carrying the trait. Initially,*

$$X \sim \mathcal{B}(100, 0.04).$$

*Since*

$$100 > 30, \qquad 0.04 < 0.1, \qquad 100 \times 0.04 = 4 < 5,$$

*we can use the Poisson approximation:*

$$X \approx \mathcal{P}(4).$$

*The probability that exactly* $8$ *individuals carry the trait is*

$$P(X = 8) \approx e^{-4}\frac{4^8}{8!}.$$

*Hence,*

$$P(X = 8) \approx 0.029.$$

# 6 Continuous Random Variables

In many practical situations, the values taken by a random variable cannot be counted individually. For example, a person's weight, height, lifetime of a machine, temperature, or blood pressure may take any value inside an interval of real numbers.

Such variables are called continuous random variables and are described by a probability density function rather than a probability mass function.

Let $X$ be a random variable whose possible values belong to an interval of $\mathbb{R}$.
The random variable $X$ is called a **continuous random variable** if its probability distribution is described by a density function $f(x)$.

The probability density function $f$ must satisfy

$$f(x) \geq 0, \qquad \forall x \in \mathbb{R},$$



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and

$$\int_{-\infty}^{+\infty} f(x)\,dx = 1.$$

The total area under the density curve is therefore equal to one.

## 6.1 Expectation

The expectation (or mean) of a continuous random variable $X$ is

$$E(X) = \int_{-\infty}^{+\infty} x\, f(x)\,dx.$$

## 6.2 Variance

The variance of a continuous random variable is

$$V(X) = E(X^2) - [E(X)]^2,$$

where

$$E(X^2) = \int_{-\infty}^{+\infty} x^2 f(x)\,dx.$$

**Important Formulas**

For a continuous random variable $X$,

$$E(X) = \int_{-\infty}^{+\infty} x\, f(x)\,dx$$

and

$$V(X) = E(X^2) - [E(X)]^2.$$

## 6.3 Probability on an Interval

For a continuous random variable, probabilities are obtained by computing areas under the density curve.

For any real numbers $a < b$,

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx.$$

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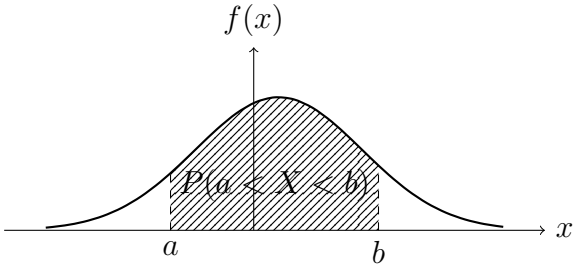



Figure 1: Probability as an area under the density curve.

## 6.4 Distribution Function

The cumulative distribution function (CDF) of $X$ is defined by

$$F(x) = P(X \leq x).$$

Using the density function, we obtain

$$F(x) = \int_{-\infty}^{x} f(u)\, du.$$

**Important Formulas**

For a continuous random variable,

$$P(X \leq a) = P(X < a)$$

and

$$P(X \geq a) = P(X > a).$$

Indeed,

$$P(X = a) = 0.$$

Since

$$P(X = a) = P(a \leq X \leq a) = \int_{a}^{a} f(x)\, dx = 0,$$

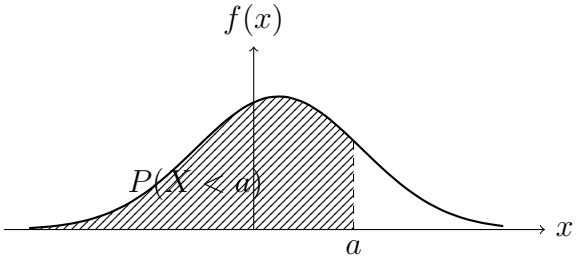



Figure 2: Probability $P(X < a)$ as an area under the density curve.

the probability of observing exactly one value is always zero for a continuous random variable.

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## 6.5 Normal Distribution

The normal distribution, also called the Gaussian or Laplace-Gauss distribution, is one of the most important probability distributions in statistics and data science. Many natural phenomena such as heights, weights, measurement errors, and biological variables can often be approximated by a normal distribution.

A continuous random variable $X$ is said to follow a **normal distribution** with mean $\mu$ and variance $\sigma^2$ if its probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right), \qquad x \in \mathbb{R}.$$

We write

$$X \sim \mathcal{N}(\mu, \sigma^2).$$

**Important Formulas**

If

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

then

$$E(X) = \mu$$

and

$$V(X) = \sigma^2.$$

The graph of the normal density has the following properties:

- The curve is symmetric with respect to $x = \mu$.
- The total area under the curve is equal to 1.
- Half of the area lies on each side of the mean:

$$P(X \leq \mu) = P(X \geq \mu) = 0.5.$$

The parameter $\mu$ determines the center of the distribution, while the parameter $\sigma^2$ controls its spread.

## 6.6 Standard Normal Distribution

A normal random variable having mean 0 and variance 1 is called a **standard normal random variable**.

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If

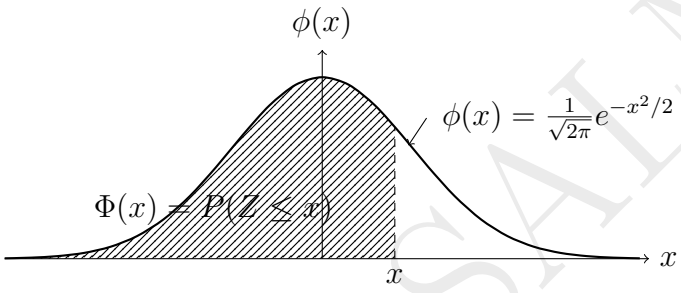
$$\mu = 0 \qquad \text{and} \qquad \sigma^2 = 1,$$

then

$$X \sim \mathcal{N}(0, 1).$$

Its density function is

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \qquad x \in \mathbb{R}.$$




$$Z \sim \mathcal{N}(0, 1)$$

Figure 3: Standard normal distribution.

The cumulative distribution function of the standard normal distribution is denoted by

$$\Phi(x).$$

It is defined by

$$\Phi(x) = P(X \leq x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, du.$$

The values of $\Phi(x)$ are obtained from the standard normal distribution table.

**Important Formulas**

For a standard normal random variable,

$$\Phi(x) = P(X \leq x).$$

Moreover,

$$P(X > x) = 1 - \Phi(x).$$

For any $a < b$,

$$P(a < X < b) = \Phi(b) - \Phi(a).$$

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## 6.7 Chi-Square Distribution

The chi-square distribution is constructed from standard normal random variables. It appears naturally when we study sums of squared standardized quantities.

Let

$$Z_1, Z_2, \ldots, Z_n$$

be $n$ independent standard normal random variables, where

$$Z_i \sim \mathcal{N}(0,1).$$

We define

$$X = Z_1^2 + Z_2^2 + \cdots + Z_n^2.$$

Then $X$ follows a **chi-square distribution** with $n$ degrees of freedom. We write

$$X \sim \chi^2(n).$$

A random variable $X$ follows a chi-square distribution with $n$ degrees of freedom if its density is

$$f(x) = \frac{1}{2^{n/2}\Gamma(n/2)} x^{\frac{n}{2}-1} e^{-x/2}, \qquad x > 0.$$

For $x \leq 0$,

$$f(x) = 0.$$

Important Formulas

The Gamma function is defined by

$$\Gamma(a) = \int_0^{+\infty} t^{a-1} e^{-t}\, dt.$$

For integer values,

$$\Gamma(n) = (n-1)!.$$

Moreover,

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

**Important Formulas**

If $X \sim \chi^2(n)$, then

$$E(X) = n$$

and

$$V(X) = 2n.$$

The chi-square distribution is not symmetric. It is defined only for positive values and becomes more spread out when the number of degrees of freedom increases.

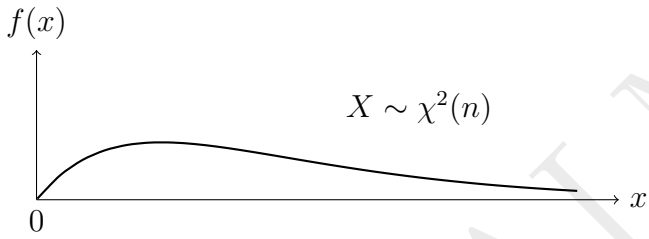



Figure 4: Shape of a chi-square density.

## 6.8 Student Distribution

The Student distribution, also called the $t$-distribution, appears when we standardize a normal random variable using an estimated standard deviation instead of the true standard deviation.

Let

$$Z \sim \mathcal{N}(0, 1)$$

and let

$$Y \sim \chi^2(n).$$

Assume that $Z$ and $Y$ are independent. We define

$$T = \frac{Z}{\sqrt{Y/n}}.$$

Then $T$ follows a **Student distribution** with $n$ degrees of freedom. We write

$$T \sim t(n).$$

A random variable $T$ follows a Student distribution with $n$ degrees of freedom if its density is

$$f(t) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\Gamma\left(\frac{n}{2}\right)} \left(1 + \frac{t^2}{n}\right)^{-\frac{n+1}{2}}, \qquad t \in \mathbb{R}.$$

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**Important Formulas**

If $T \sim t(n)$, then the density is symmetric around 0.
When $n$ becomes large,

$$t(n) \approx \mathcal{N}(0,1).$$

The Student distribution has heavier tails than the standard normal distribution. This means that extreme values are more likely under a Student distribution than under a normal distribution.

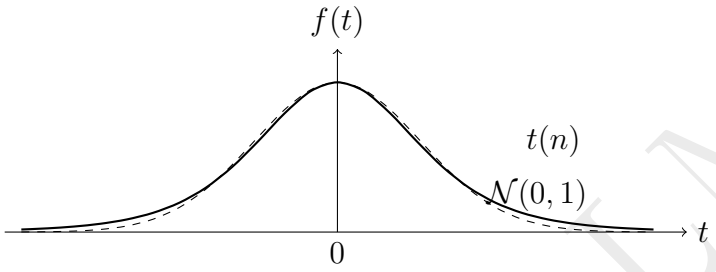



Figure 5: Student density compared with the standard normal density.

# 7 Central Limit Theorem and Continuity Correction

One of the most important results in probability and statistics is the Central Limit Theorem (CLT). It explains why the normal distribution appears so frequently in practical applications.

Even when the original population is not normally distributed, the distribution of the sample mean becomes approximately normal when the sample size is sufficiently large.

The Central Limit Theorem is the foundation of statistical inference, confidence intervals, hypothesis testing, and many machine learning algorithms.

## 7.1 Central Limit Theorem

Let

$$X_1, X_2, \ldots, X_n$$

be independent and identically distributed random variables such that

$$E(X_i) = \mu$$

and

$$V(X_i) = \sigma^2.$$

Define the sample mean

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

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**Central Limit Theorem**

When the sample size $n$ becomes large,

$$\overline{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

Equivalently,

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \approx \mathcal{N}(0, 1).$$

**Important Formulas**

In practice, the approximation is usually considered satisfactory whenever

$$n \geq 30.$$

Therefore, for large samples,

$$\overline{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

**Remark.**

If the population itself follows a normal distribution,

$$X_i \sim \mathcal{N}(\mu, \sigma^2),$$

then

$$\overline{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

for every sample size $n$, even when $n < 30$.

## 7.2 Normal Approximation of a Binomial Distribution

$$X \sim \mathcal{B}(n, p)$$

If

$$n \geq 30$$

and

$$np(1-p) \geq 5,$$

then

$$X \approx \mathcal{N}\Big(np, np(1-p)\Big).$$

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## 7.3 Normal Approximation of a Poisson Distribution

If

$$X \sim \mathcal{P}(\lambda)$$

and

$$\lambda \geq 25,$$

then

$$X \approx \mathcal{N}(\lambda, \lambda).$$

## 7.4 Continuity Correction

When a discrete distribution is approximated by a continuous normal distribution, a correction called the continuity correction must be applied.

The idea is to replace a single integer value by an interval of width one centered around that integer.

**Important Formulas**

$$P(X = a) \approx P(a - 0.5 < X < a + 0.5)$$

$$P(X \leq a) \approx P(X < a + 0.5)$$

$$P(X < a) \approx P(X < a - 0.5)$$

$$P(X \geq a) \approx P(X > a - 0.5)$$

$$P(X > a) \approx P(X > a + 0.5)$$

**Example 7.1.**

*The average age of cars in a city is*

$$\mu = 20 \text{ years}$$

*with standard deviation*

$$\sigma = 5 \text{ years}.$$

*A random sample of*

$$n = 36$$

*cars is selected.*

*Let $\overline{X}$ denote the sample mean age. Find the probability that the average age of the selected cars is between 18 and 22 years.*

***Solution***

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*Since*

$$n = 36 > 30,$$

*the Central Limit Theorem implies that*

$$\overline{X} \approx \mathcal{N}\left(20, \frac{5^2}{36}\right).$$

*Hence*

$$P(18 < \overline{X} < 22) = P\left(\frac{18-20}{5/\sqrt{36}} < Z < \frac{22-20}{5/\sqrt{36}}\right),$$

*where*

$$Z \sim \mathcal{N}(0,1).$$

*Therefore*

$$P(18 < \overline{X} < 22) = P(-2.4 < Z < 2.4).$$

*Using the standard normal table,*

$$P(-2.4 < Z < 2.4) = \Phi(2.4) - \Phi(-2.4).$$

*Since*

$$\Phi(2.4) = 0.9918$$

*and*

$$\Phi(-2.4) = 0.0082,$$

*we obtain*

$$P(18 < \overline{X} < 22) = 0.9918 - 0.0082 = 0.9836.$$

*Thus,*

$$\boxed{P(18 < \overline{X} < 22) = 0.9836}$$

*which means that there is approximately a* 98.36% *probability that the sample mean age lies between* 18 *and* 22 *years.*

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